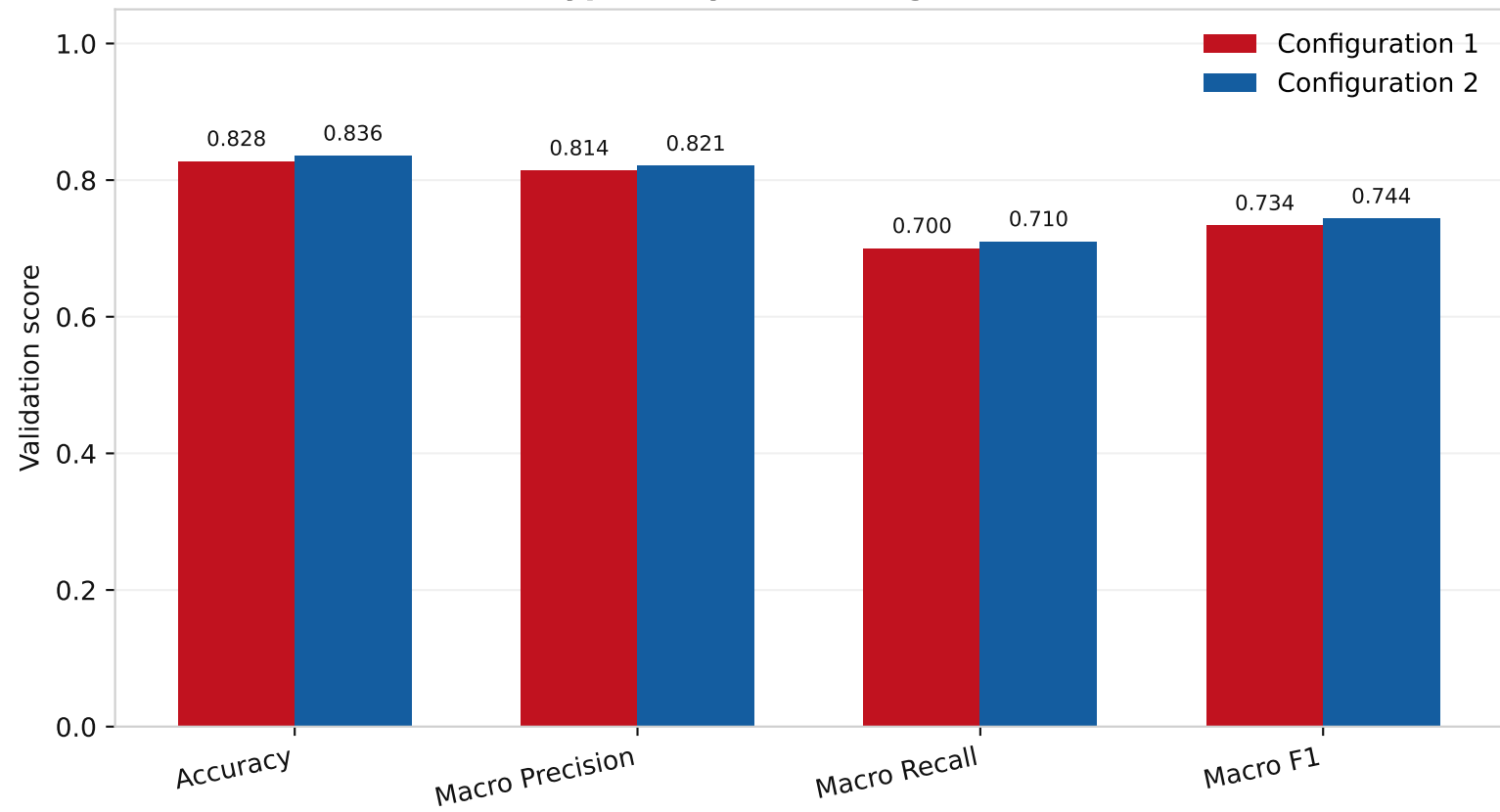
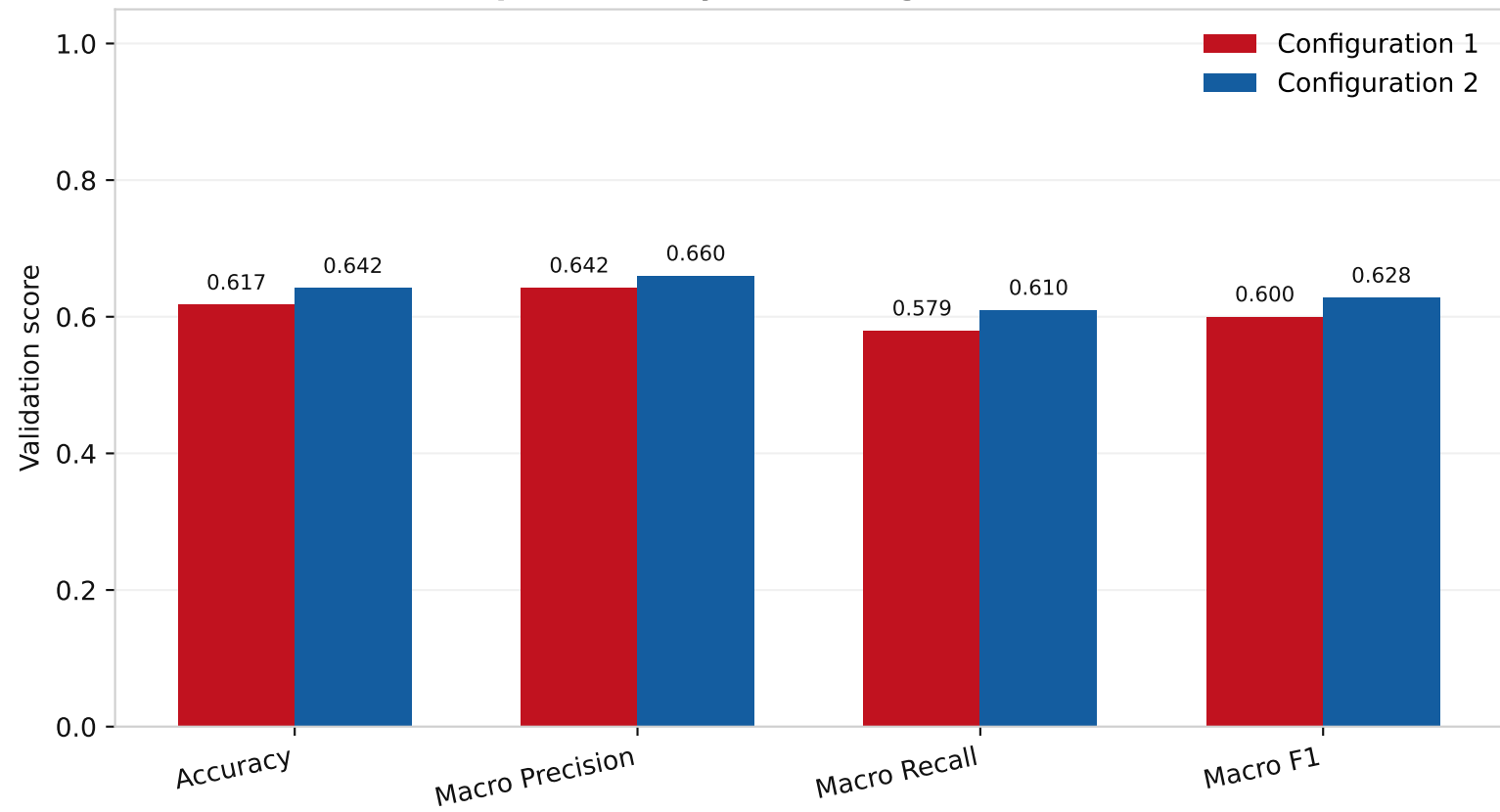


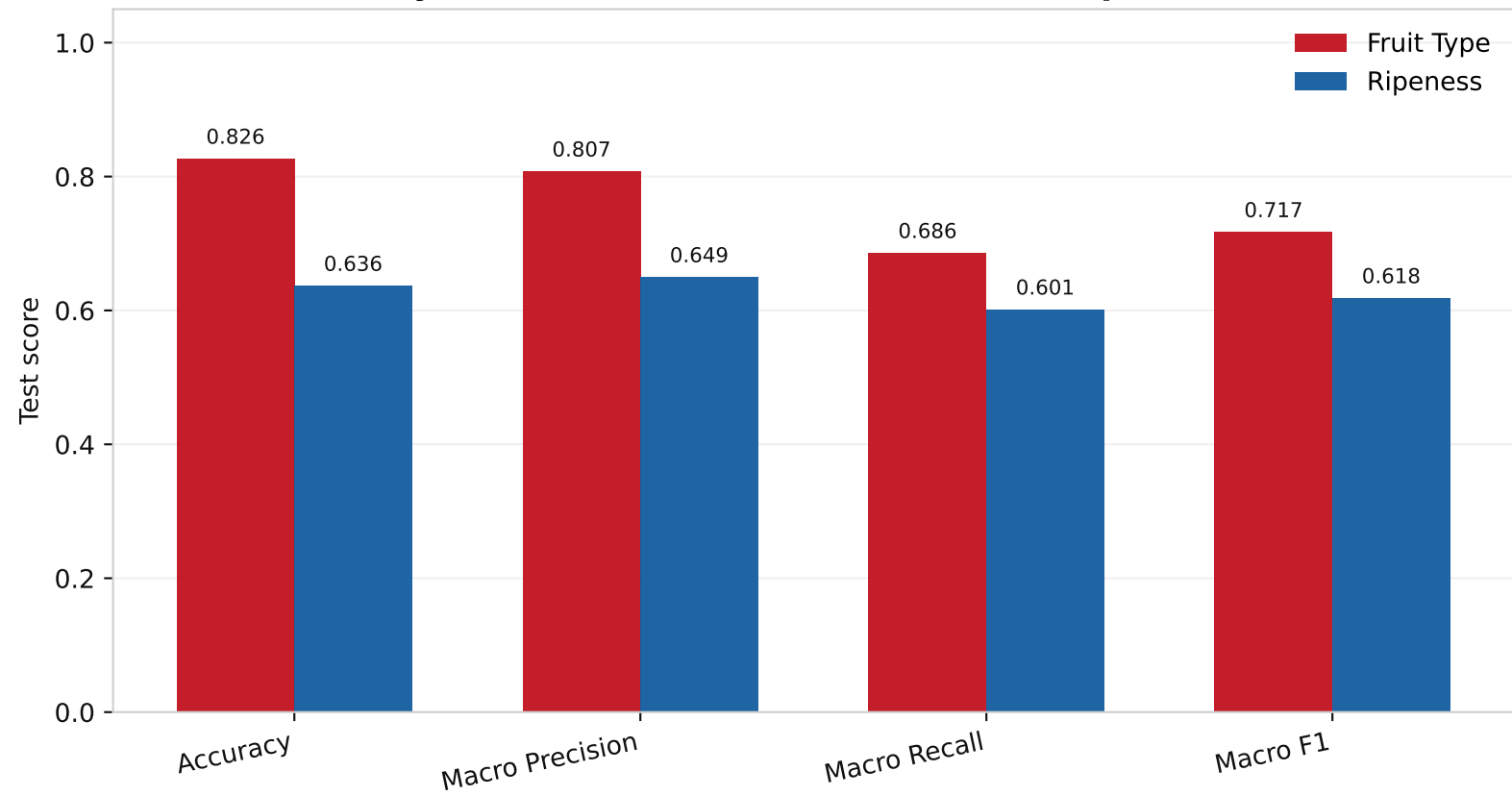
Fruit Type — Hybrid Tuning Performance



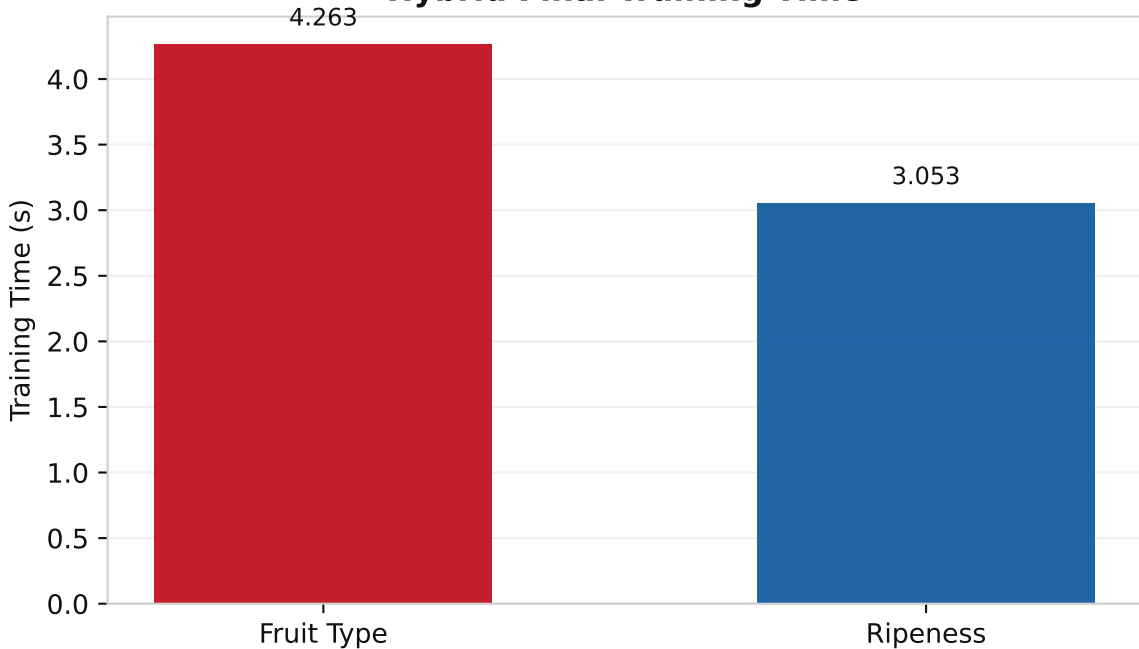
Ripeness — Hybrid Tuning Performance



Hybrid Final Classification Metric Comparison

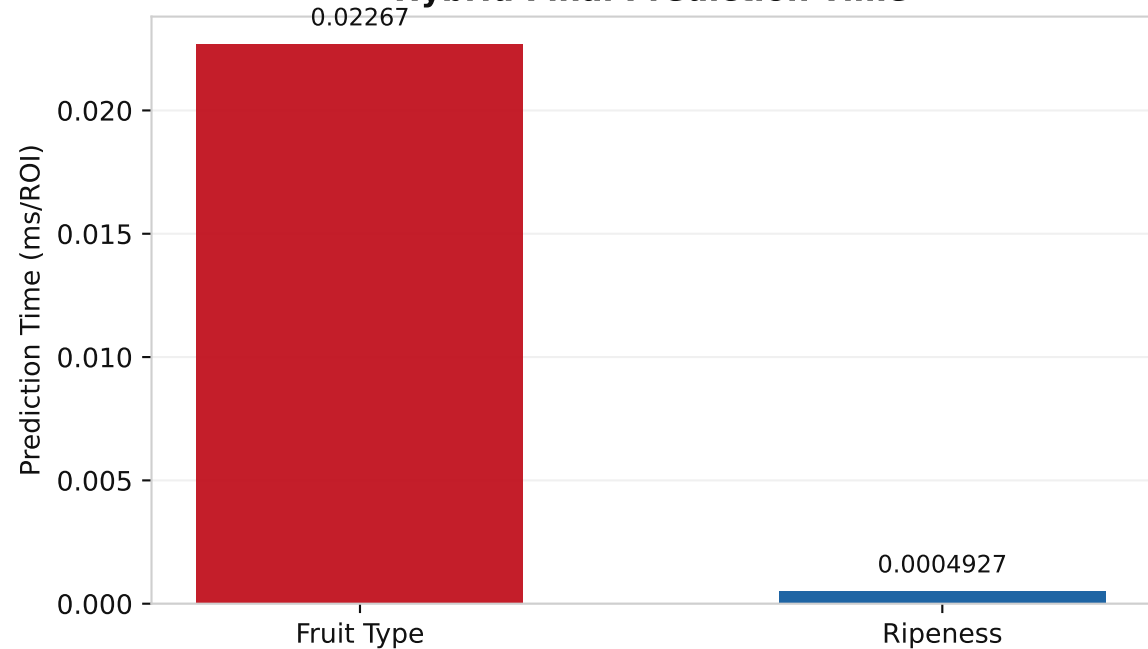


Hybrid Final Training Time



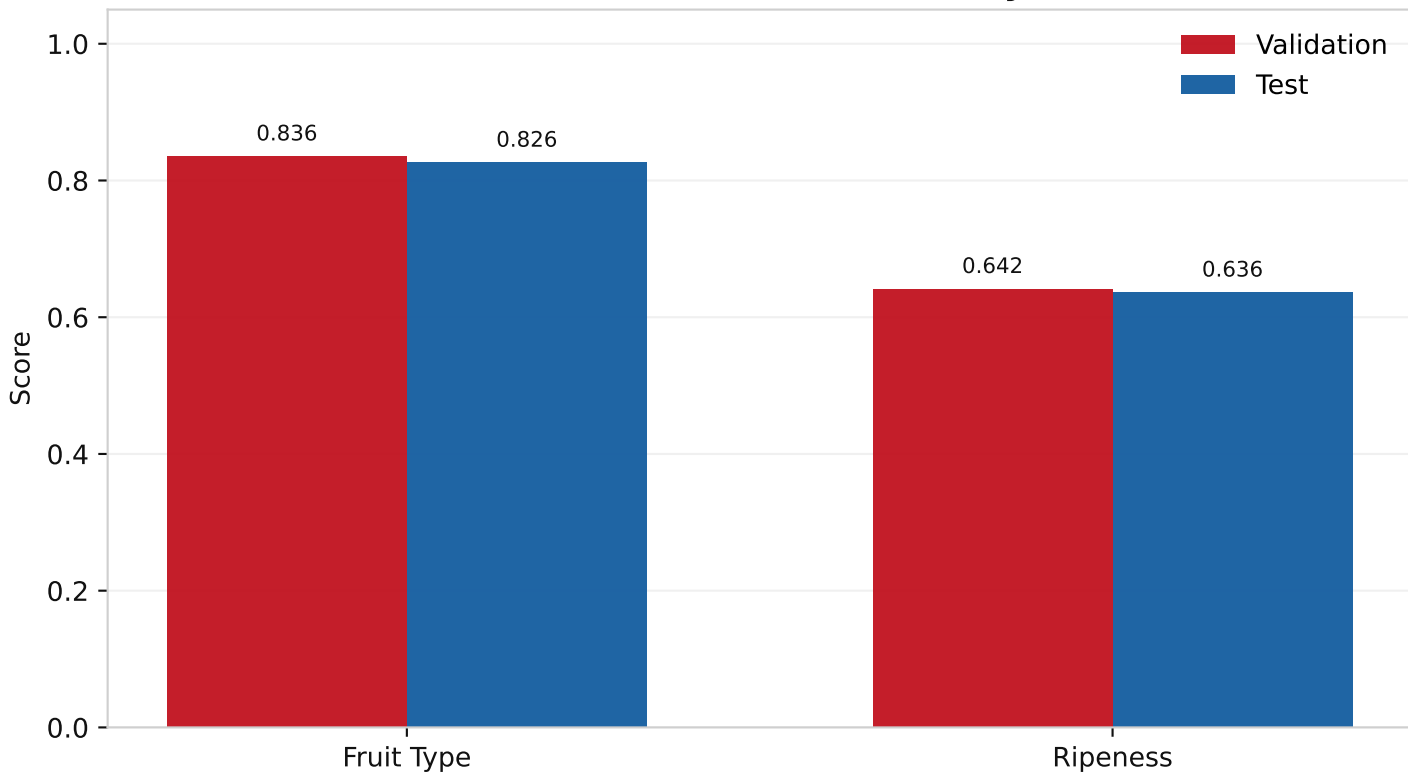
Lower is better

Hybrid Final Prediction Time

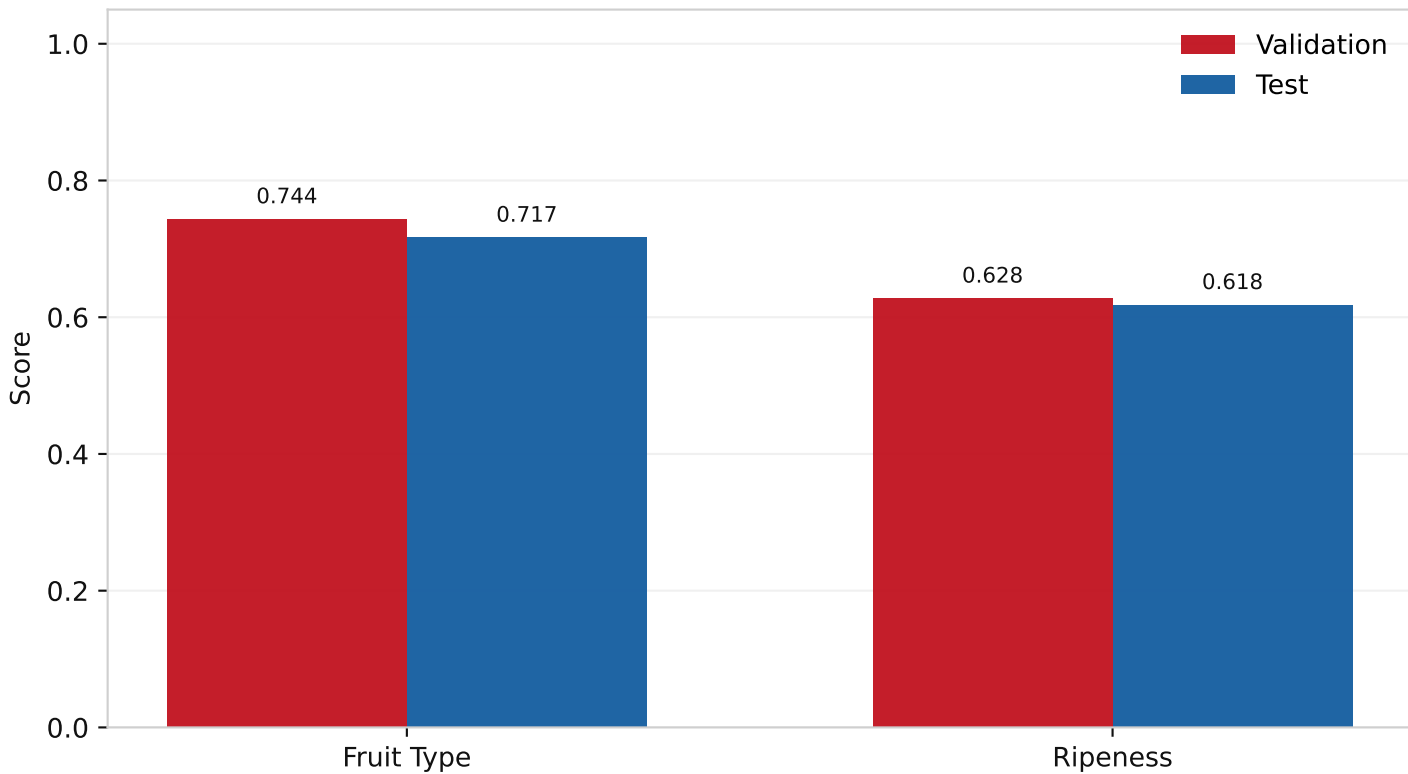


Lower is better

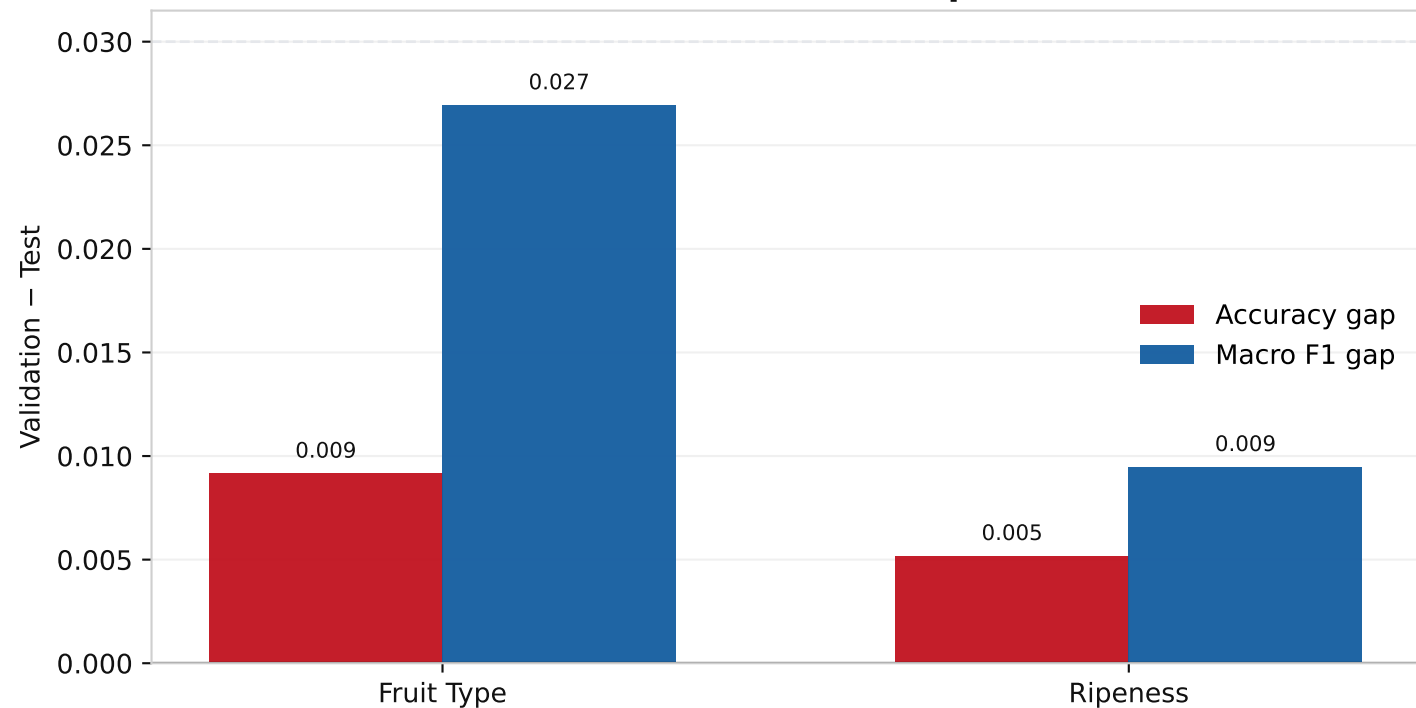
Validation vs Test Accuracy



Validation vs Test Macro F1

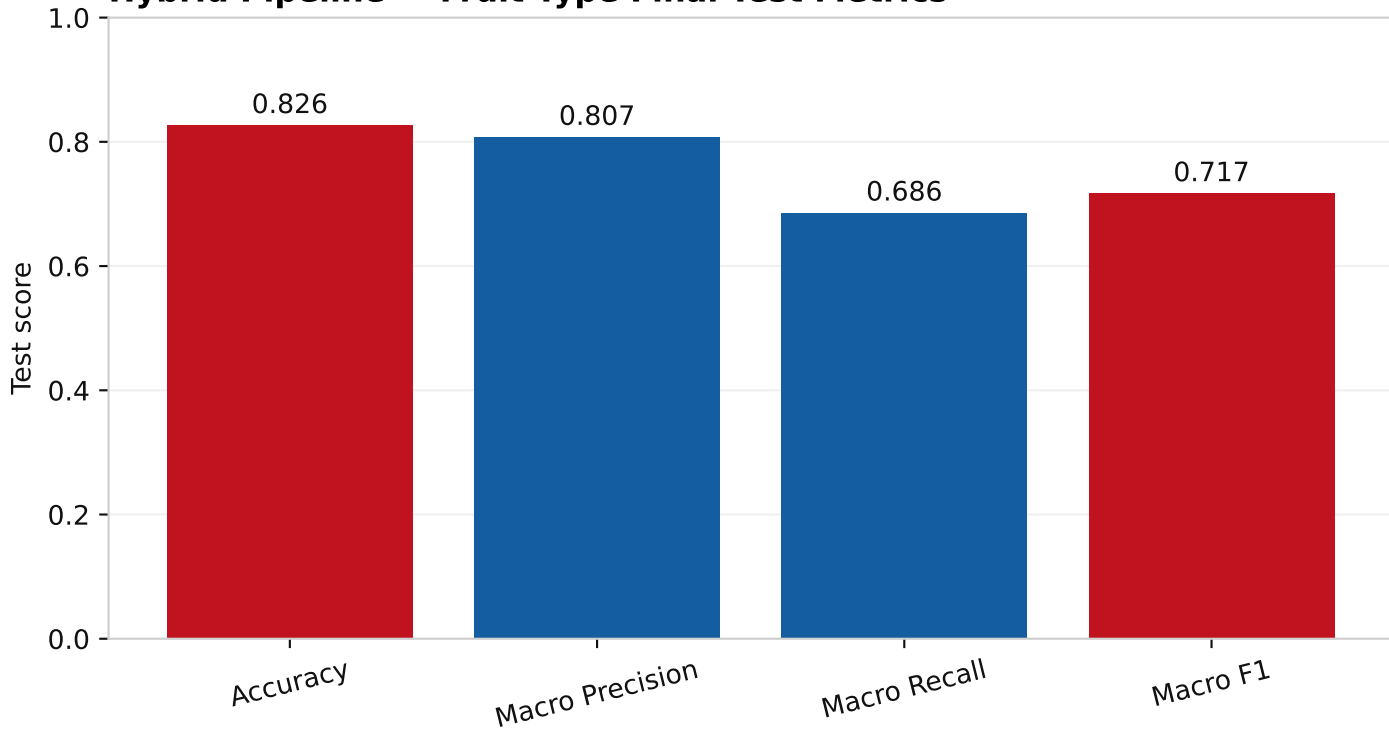


Generalization Gap



A small gap indicates similar Validation and Test performance

Hybrid Pipeline — Fruit Type Final Test Metrics



Hybrid Fruit Type Classification Radar

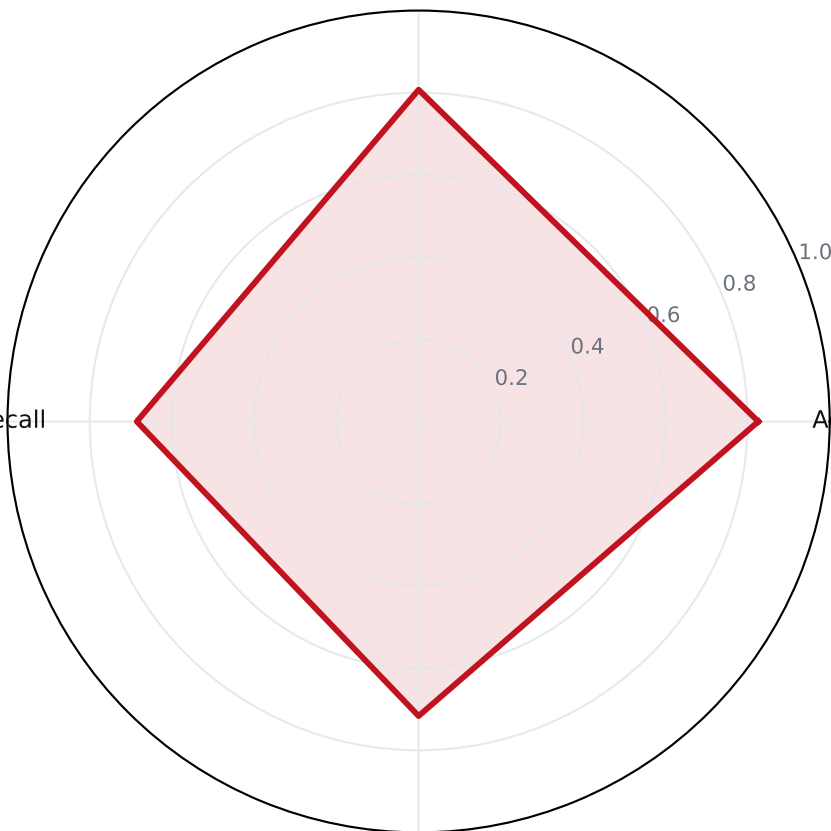
Macro Recall

Macro Precision

Hybrid Fruit Type

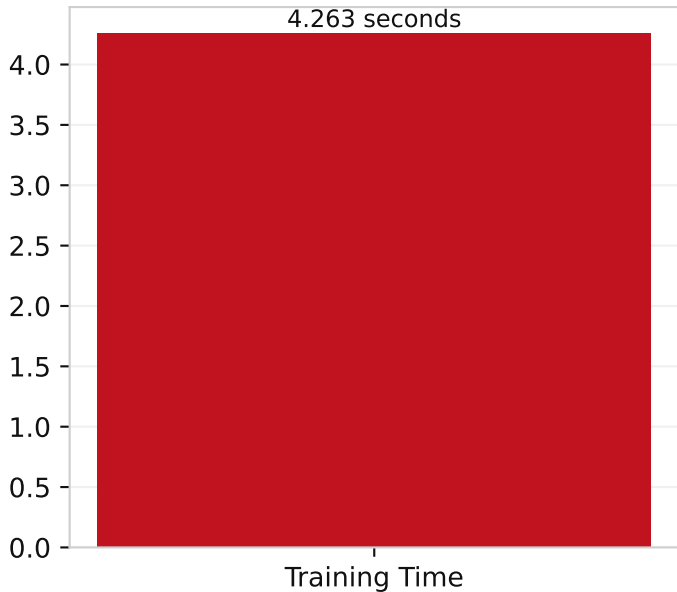
Accuracy

Macro F1

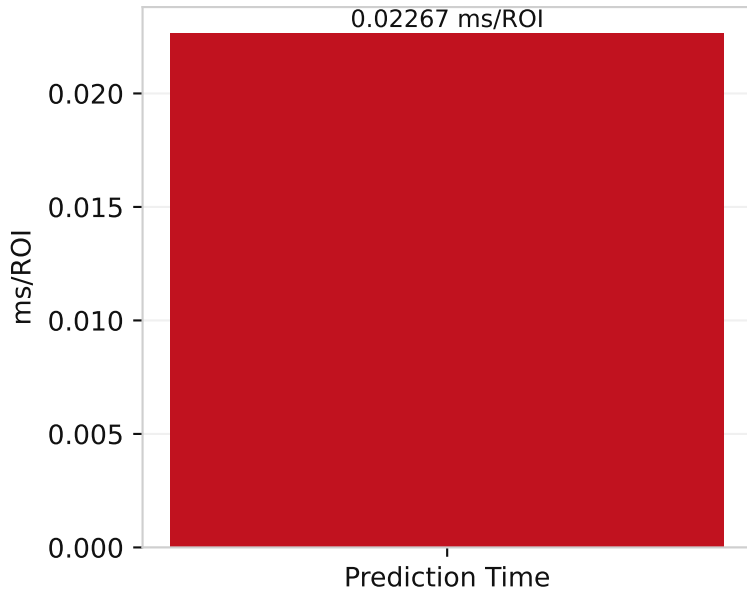


Hybrid Fruit Type Efficiency

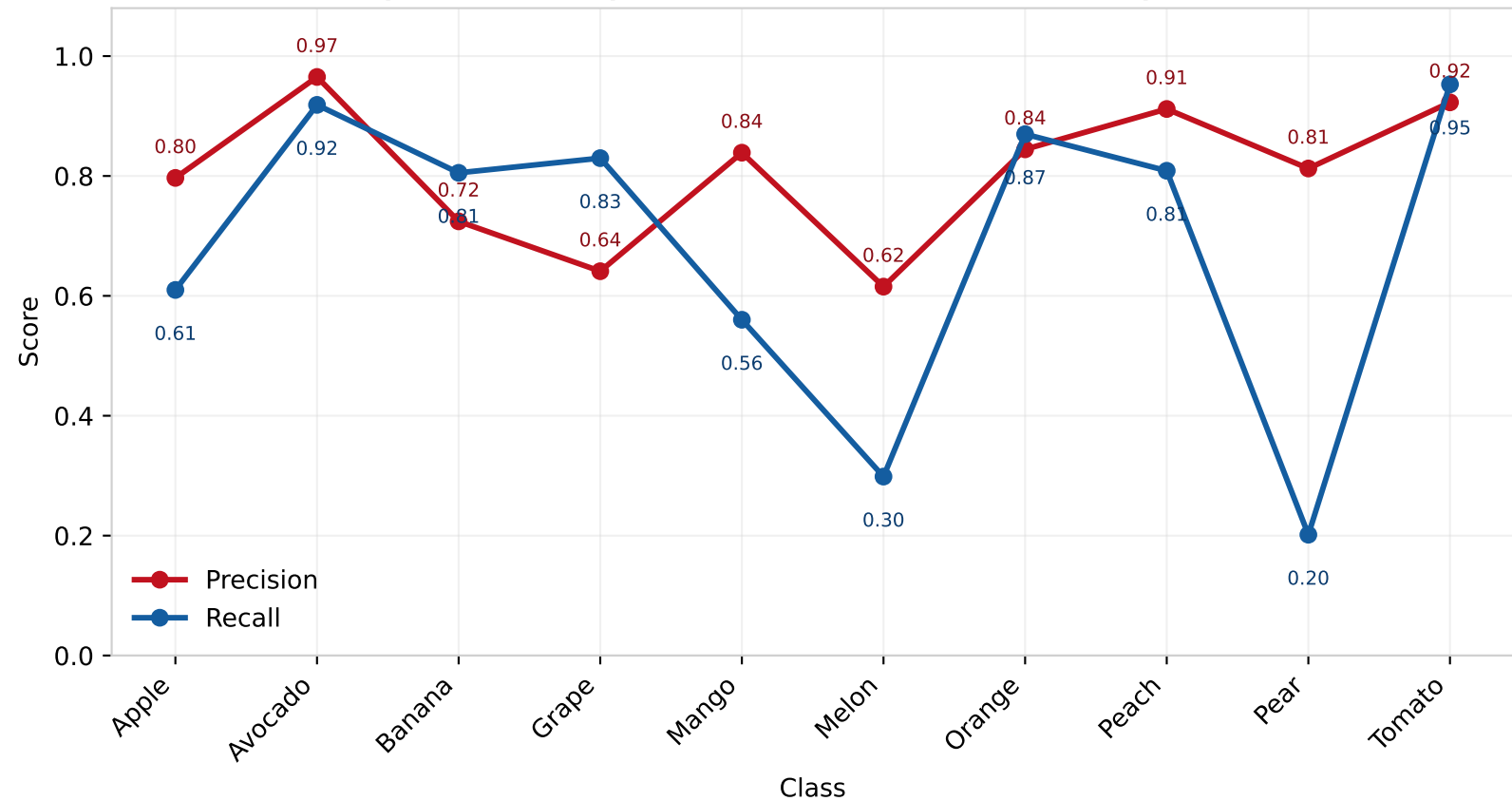
Fruit Type Training Time



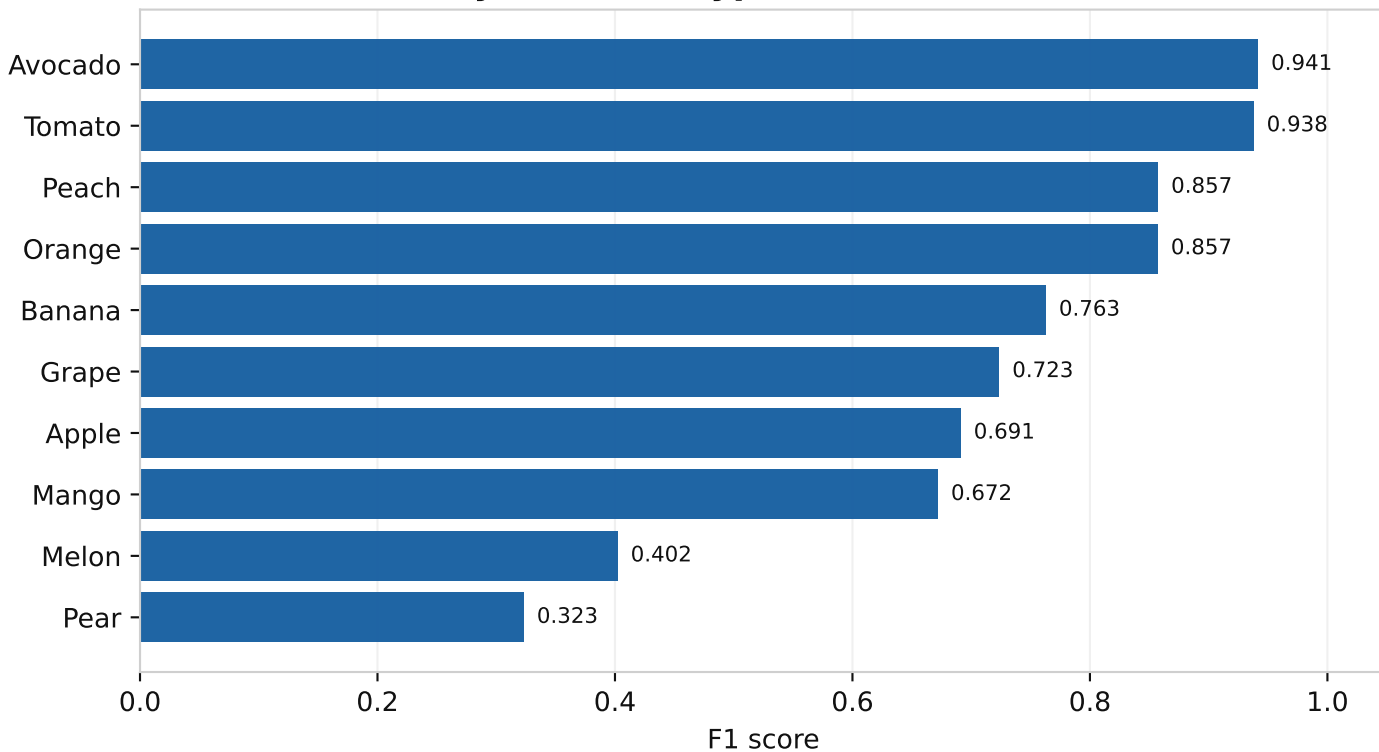
Fruit Type Prediction Time



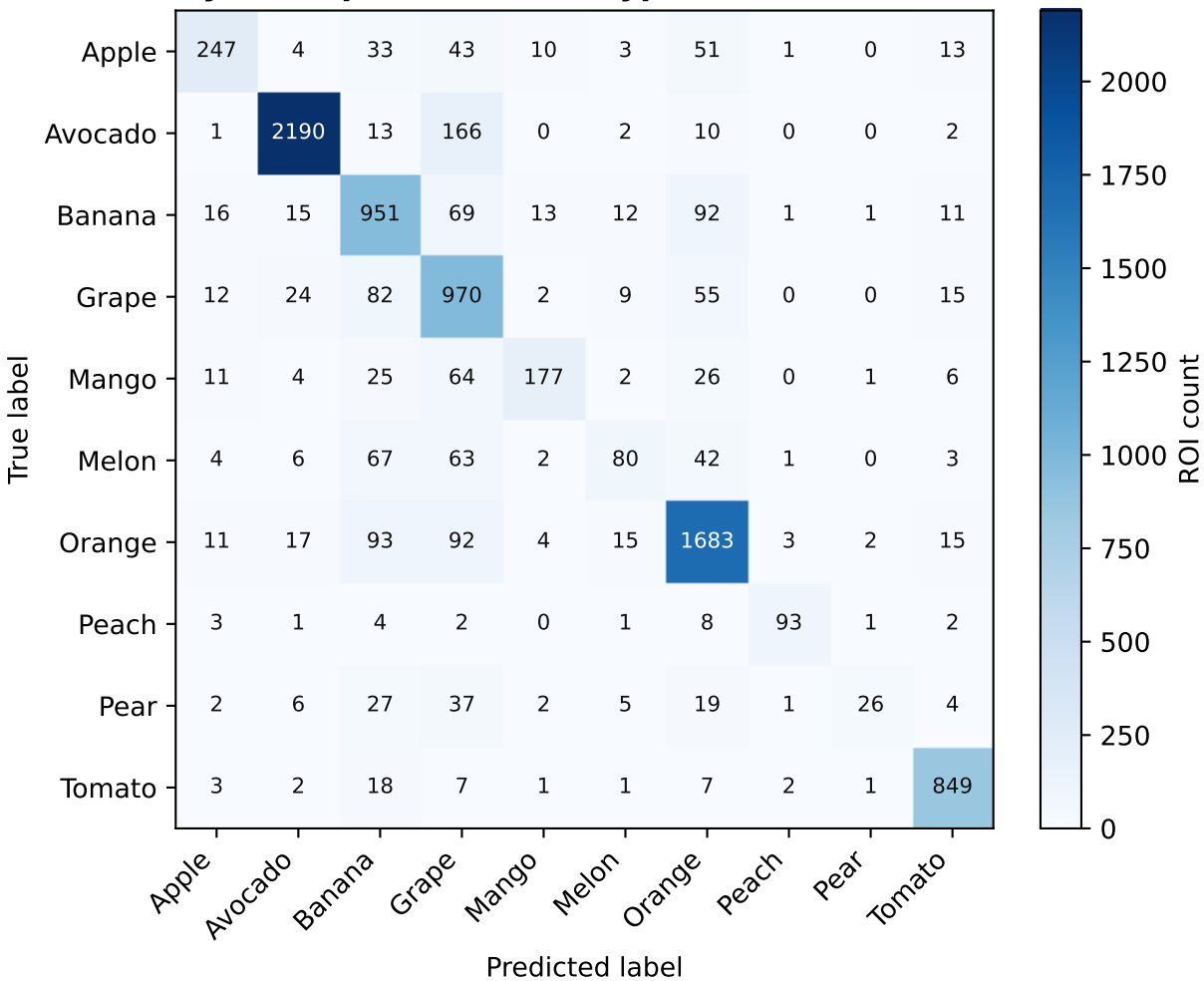
Hybrid Fruit Type — Precision and Recall by Class



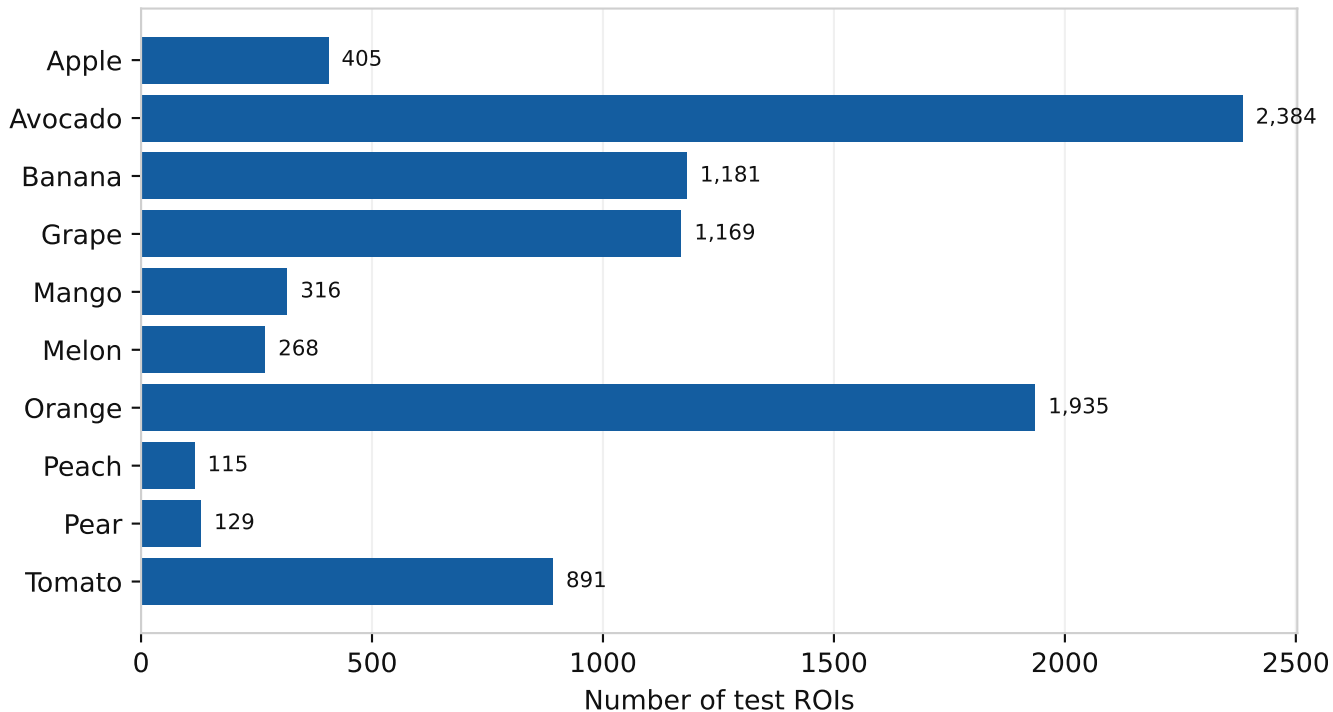
Hybrid Fruit Type — Per-Class F1



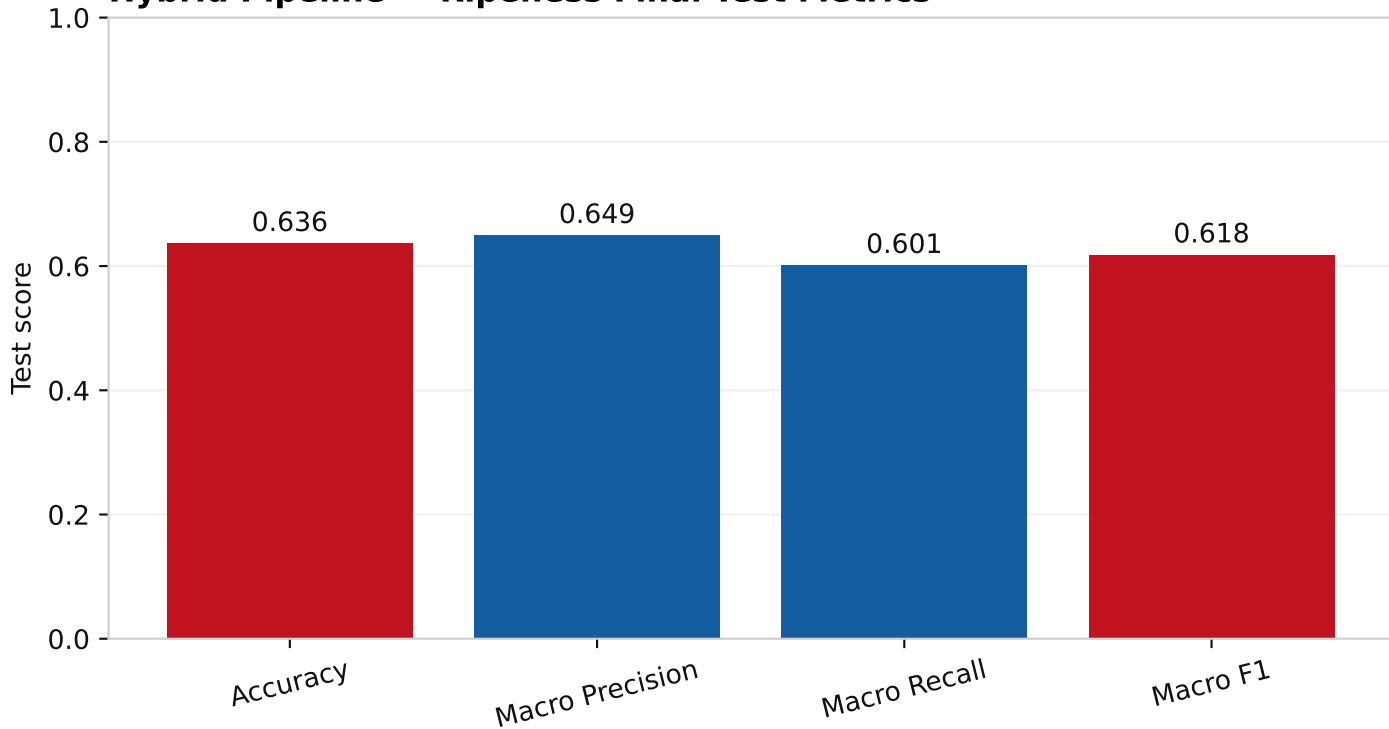
Hybrid Pipeline — Fruit Type Confusion Matrix



Hybrid Fruit Type Test Support



Hybrid Pipeline — Ripeness Final Test Metrics



Hybrid Ripeness Classification Radar

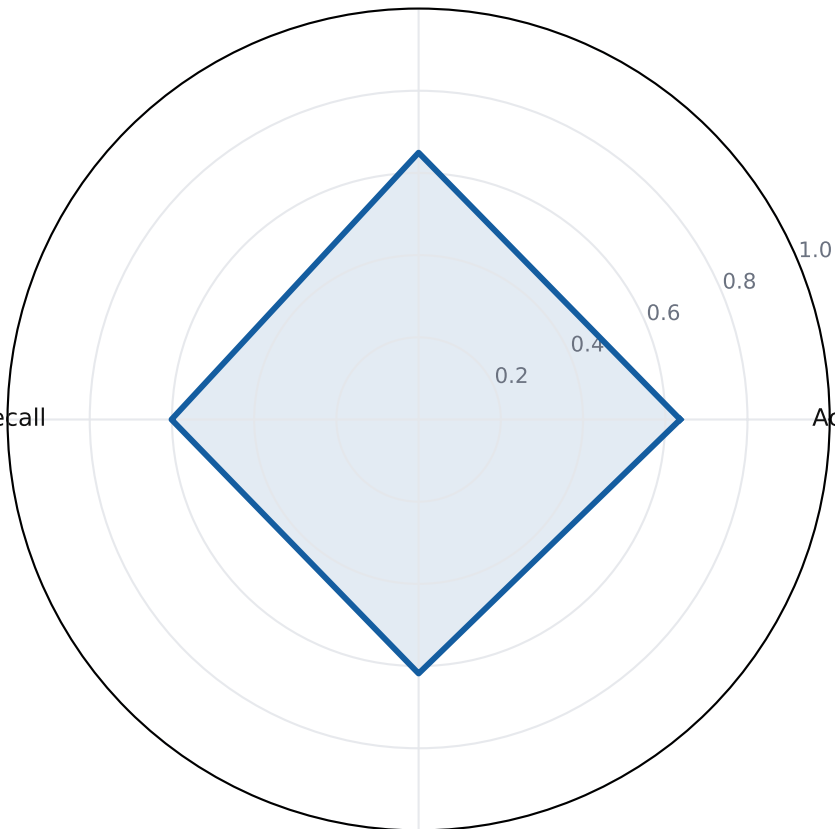
Macro Recall

Macro Precision

Hybrid Ripeness

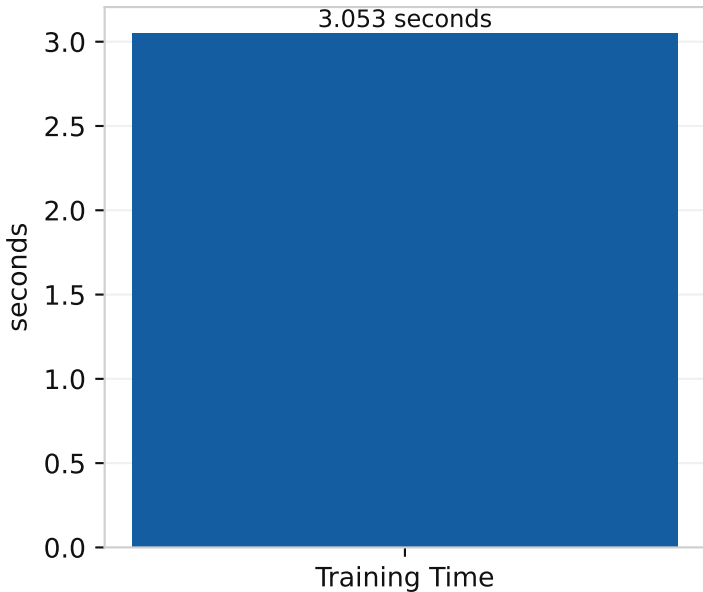
Accuracy

Macro F1

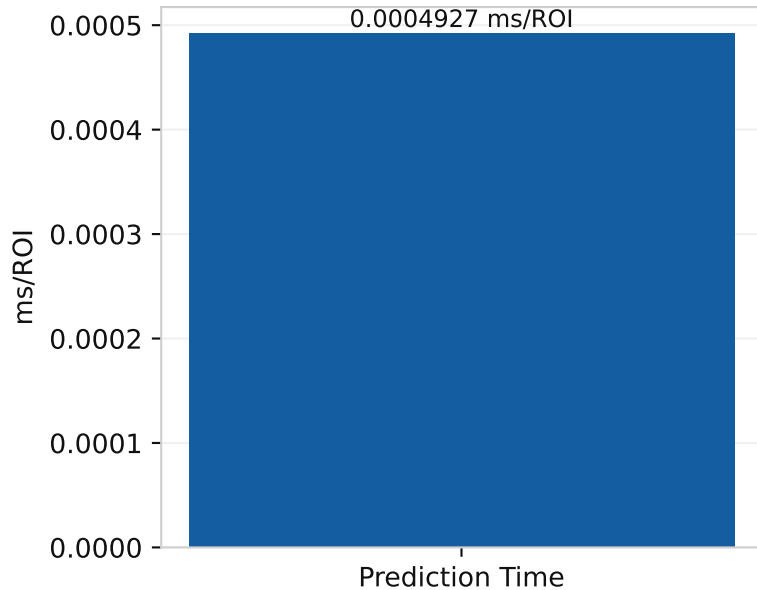


Hybrid Ripeness Efficiency

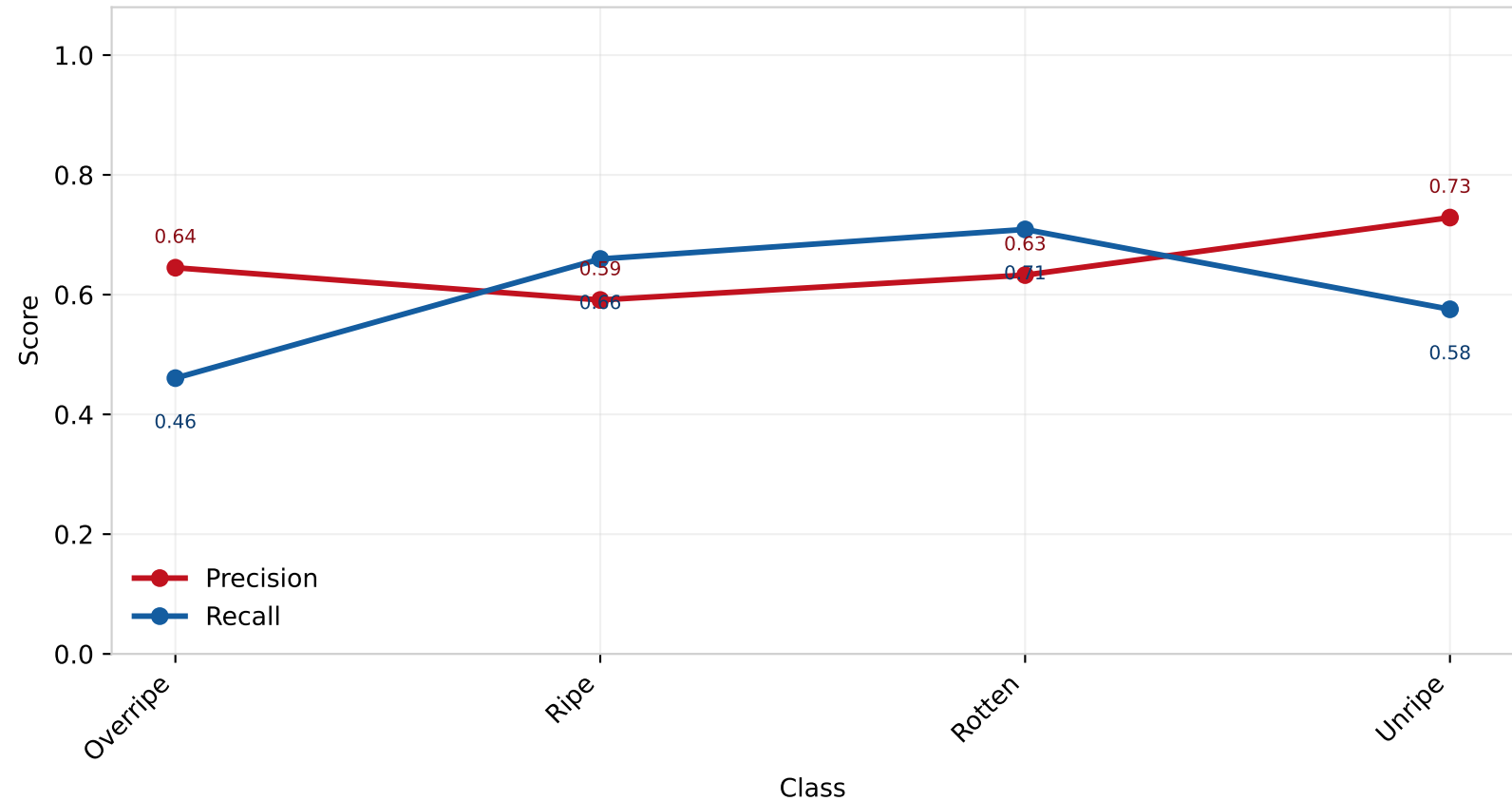
Ripeness Training Time



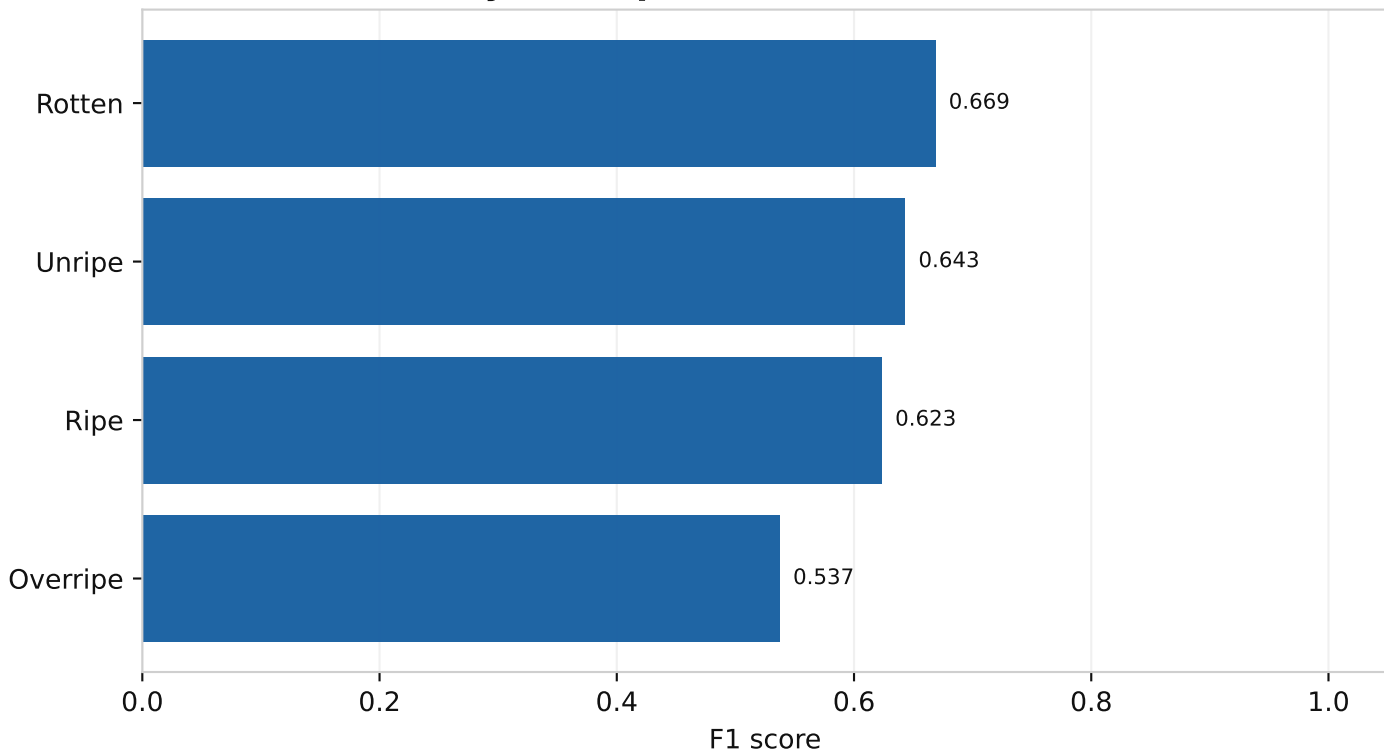
Ripeness Prediction Time



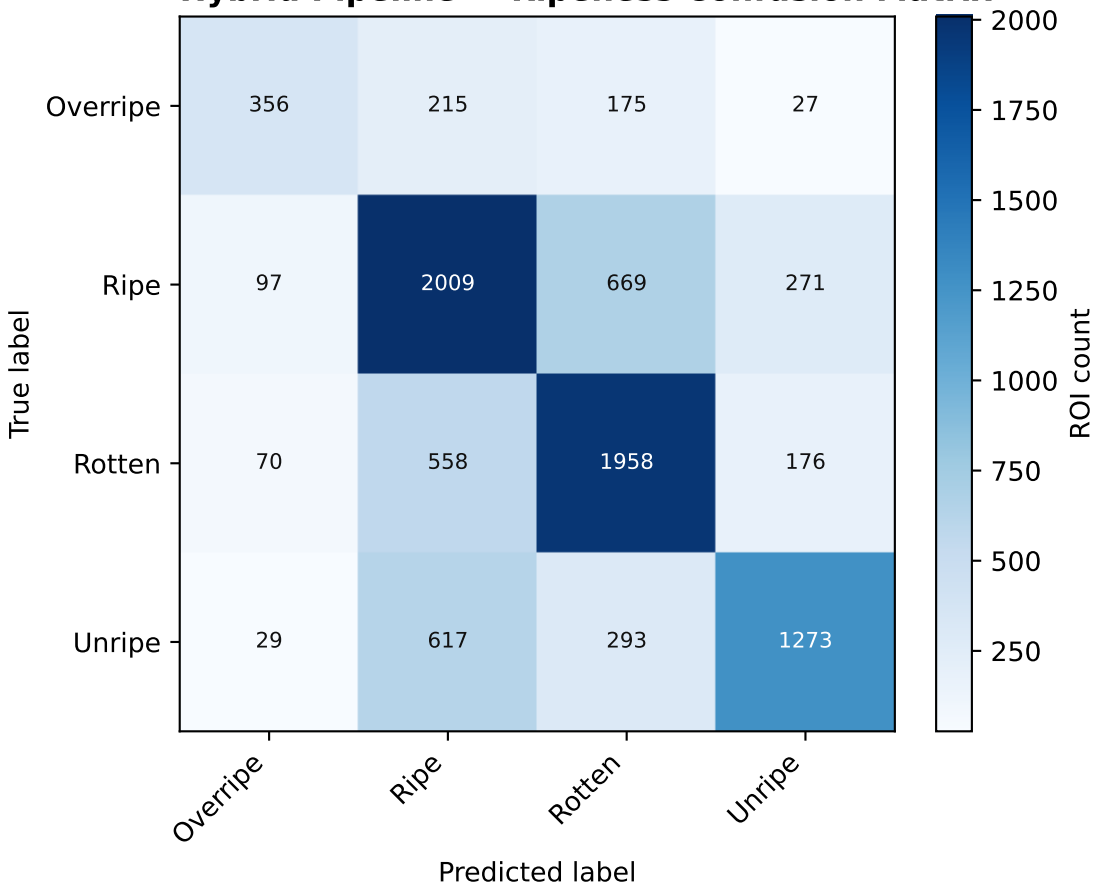
Hybrid Ripeness — Precision and Recall by Class



Hybrid Ripeness — Per-Class F1



Hybrid Pipeline — Ripeness Confusion Matrix



Hybrid Ripeness Test Support

